SALES FORECASTING & DEMAND INTELLIGENCE

3-MO. FORECAST (END)	RECOMMENDED MODEL	TOP CATEGORY	ANOMALOUS WEEKS
$107,211	**SARIMA**	**Technology**	**7 flagged**

1. Executive Summary

We built a system that forecasts product demand up to three months ahead by learning from four years of daily sales history, then layered on two additional capabilities: automatic detection of unusual sales weeks, and grouping of products by demand behavior so stocking policy can be tailored rather than one-size-fits-all. The headline number: total sales are projected to grow from $73,694 next month to $107,211 by month three, a steady upward trajectory rather than a one-off spike. Three things stand out from the analysis. First, Technology is our clear revenue engine and East is our most dependable growth region — both are safe targets for increased investment. Second, demand is sharply seasonal, with November and December reliably outperforming every other month, which means procurement needs to lead the season by three to four weeks rather than react to it. Third, not all products behave the same way — some are steady and predictable, others are high-value but erratic, and a few are small today but growing fast — so a single inventory policy applied company-wide is leaving money on the table in at least two of the four segments identified below.

2. Key Findings from Exploratory Analysis

- Highest-revenue category: Technology, $827,456 total over 4 years — roughly 14% ahead of Furniture ($728,659) and 17% ahead of Office Supplies ($705,422), making it the single most important category to keep in stock
- Most consistent regional growth: East — its year-over-year growth is far more stable than other regions (variability score 0.10, vs. 1.20 for West and 1.95+ for Central/South). A lower score means fewer surprises year to year, so East is the region we can plan inventory commitments against with the most confidence
- Average shipping time: 3.96 days company-wide, and it barely varies by region — East ships fastest at 3.91 days, Central slowest at 4.07 days. Because the spread is under half a day, shipping speed is not currently a meaningful lever for improving customer experience or cutting costs
- Seasonality: sales consistently spike every November and December across all 4 years, with a smaller recurring peak in September. This is a structural, repeating pattern rather than a one-time event, which means it can and should be planned for on a fixed annual calendar

3. 3-Month Sales Forecast

We tested three fundamentally different forecasting techniques head-to-head — SARIMA (a classical statistical model), Prophet (Meta's production forecasting tool), and XGBoost (a machine-learning approach) — and evaluated each on how far its predictions strayed from actual held-out sales. Three metrics were used:

MAE (average dollar error per month), RMSE (which penalizes larger misses more heavily), and MAPE (average error as a percentage of actual sales, easiest to compare across scale). SARIMA won on all three, meaning it was both the most accurate on average and the least prone to a single large miss — the reason it is recommended for production forecasting going forward. Using SARIMA, we project sales of $73,694 in Month 1, $95,397 in Month 2, and $107,211 in Month 3. A margin of roughly ±$23,500 should be treated as the normal range of outcomes, not a sign of an unreliable model — all forecasts carry uncertainty, and this range reflects the natural month-to-month variability already present in four years of historical sales.

Model	MAE	RMSE	MAPE	Month 1	Month 2	Month 3
SARIMA ★	16,825.17	19,208.46	17.69%	$73,694.38	$95,396.63	$107,210.62
Prophet	20,250.79	22,318.41	21.86%	$51,318.94	$90,325.65	$90,041.08
XGBoost	18,605.67	20,520.53	19.09%	$67,994.23	$87,716.29	$66,889.15

4. Top 3 Anomalies Detected

Two independent statistical methods — Isolation Forest and a rolling Z-score check — were run separately on weekly sales, and they agreed on 92.8% of all weeks, which gives confidence that the anomalies flagged below are real signal and not a quirk of one particular method. Each of the three largest spikes lines up with a plausible business explanation rather than appearing random, which is itself a useful sanity check on data quality:

Date	Sales Value	Likely Cause
2015-03-22	$37,703.67	Largest spike on record, flagged by both detection methods — likely a large bulk or corporate order concentrated in one week
2018-12-02	$35,998.90	Early-December spike, consistent with year-end holiday purchasing
2018-11-18	$30,572.45	Mid-November spike, consistent with Black Friday / holiday season shopping

5. Product Demand Segmentation

Rather than treating all products the same, sub-categories were grouped into four behavioral segments using four measurable traits: total sales volume, year-over-year growth rate, volatility (how much sales swing month to month), and average order value. This turns a long, undifferentiated product list into four clear, actionable stocking policies:

Segment	Example Sub-Categories	Recommended Stocking Strategy
High Volume, Stable Demand	Accessories, Binders, Chairs, Phones, Storage, Tables	Maintain steady safety stock; reorder on a fixed schedule — this group generates the most total revenue of any segment
High Value, Volatile Demand	Copiers, Machines	Hold extra buffer stock; these are big-ticket items with unpredictable week-to-week swings, so plan for spikes
Low Volume, Stable Demand	Appliances, Art, Bookcases, Envelopes, Fasteners, Furnishings, Labels, Paper	Keep minimal, predictable stock levels; low risk of stockouts or overstock
Niche, Rapidly Growing	Supplies	Small revenue base today but growing fast; increase reorder frequency and watch closely for a breakout trend

6. Business Recommendations

- Prioritize Technology and the East region for stock allocation. Technology leads all categories at $827,456 in total revenue, and East shows the most reliable year-over-year growth of any region — together they represent the lowest-risk, highest-confidence targets for increased inventory investment, since both have a demonstrated, stable track record rather than a speculative one.
- Pre-position inventory 3-4 weeks ahead of November every year. Sales spike to an average of $87,540 in November and $80,370 in December, well above every other month, and this pattern has now repeated across all four years of data — meaning it is a plannable calendar event, not a guess. Ordering on the usual lead time risks arriving after the demand window has already opened.
- Rebalance working capital toward the Growing and High-Volume segments and away from the Niche segment's over-investment risk. The High Volume, Stable Demand group (Accessories, Binders, Chairs, Phones, Storage, Tables) already generates the most combined revenue of any cluster and carries the lowest volatility — it is the safest place to expand stock. The Niche, Rapidly Growing segment (Supplies) is small today but growing quickly and deserves closer, more frequent reordering rather than a large one-time capital commitment until its growth trend is confirmed over another quarter.

7. Risk & Limitation

This system learns exclusively from historical sales patterns, which means it has no way to anticipate events with no precedent in the data — a new competitor entering the market, a supply chain disruption, a pricing change, or a sharp economic shift. It should be treated as a decision-support tool that narrows down and quantifies uncertainty, not a guarantee, and forecasts should be refreshed monthly as new actual sales data comes in so the model keeps learning from the most recent conditions. One specific data-quality caution: the category-level forecast for Furniture showed an unusually large swing (+200% over 3 months), which is almost certainly an artifact of a small, noisy sample at that level of granularity rather than a genuine trend — it should be manually reviewed against actual Furniture orders before any stocking decision is made on its basis.